Improving Customer Retention and Long-Term Value at Instacart

How early shopping behavior predicts who stays — and what to do about it

Luul Shire
MSBA-655-001: Retailing Analytics | Dr. Taotao Ye
Costello College of Business, George Mason University

WHY IT MATTERS

Business Context & Problem Statement

Online grocery is high-frequency, low-margin

- Instacart doesn't own inventory or set prices — its margin depends on how often customers come back, not on any single order.
- Acquiring a new customer costs real marketing spend with no guaranteed return; early churn means that spend never pays off.
- Repeat customers create predictable demand, which makes staffing, inventory partnerships, and delivery capacity easier to plan.
- Because grocery shopping repeats weekly, every order is a fresh chance to either deepen the relationship or lose it.

$285B → $765B

Global online grocery market, 2024 projected to 2030

1,400+

Retail partners Instacart connects customers to across North America

0

Inventory or warehouses owned — Instacart's model lives or dies on repeat engagement

- **Research Question: How can Instacart use early customer behavior, basket composition, and purchasing patterns to improve retention and long-term customer value?**

DATA & METHODOLOGY

Three Million Orders, One Analytical Framework

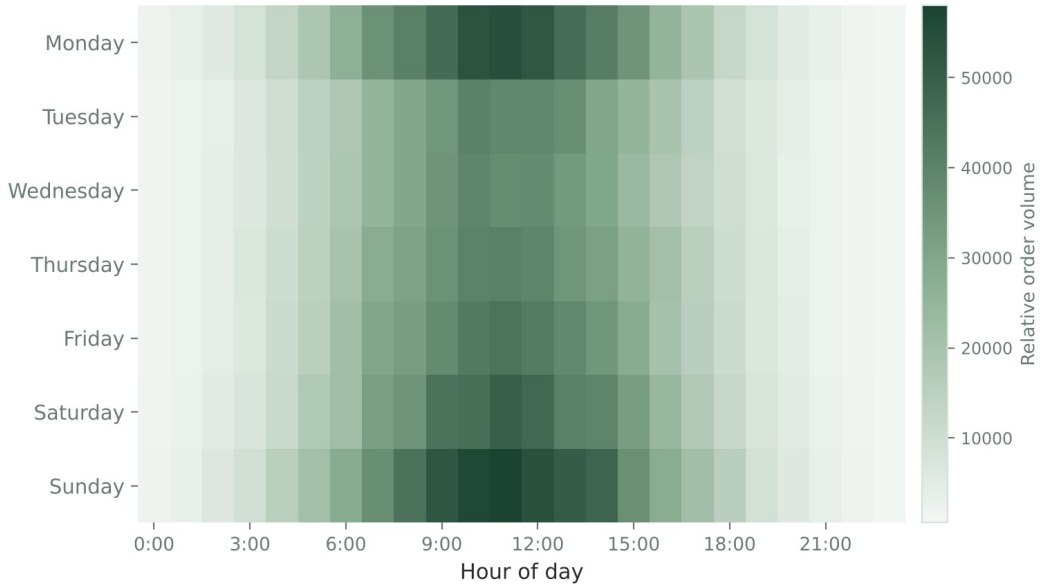
Dataset

- Instacart Market Basket Analysis dataset (public, 2017 release)
- 3M+ orders from 200,000+ users across five linked tables — orders, products, aisles, departments
- Key fields: order timing, order sequence, days since prior order, department, reorder flag

Analytical Framework

1. **Exploratory Analysis** Reorder cycles, basket size, and early disengagement signals
2. **CART Model** Predict dairy & egg purchases from basket composition
3. **K-Means Segmentation** Group customers by long-term purchasing behavior

When customers order: day × hour

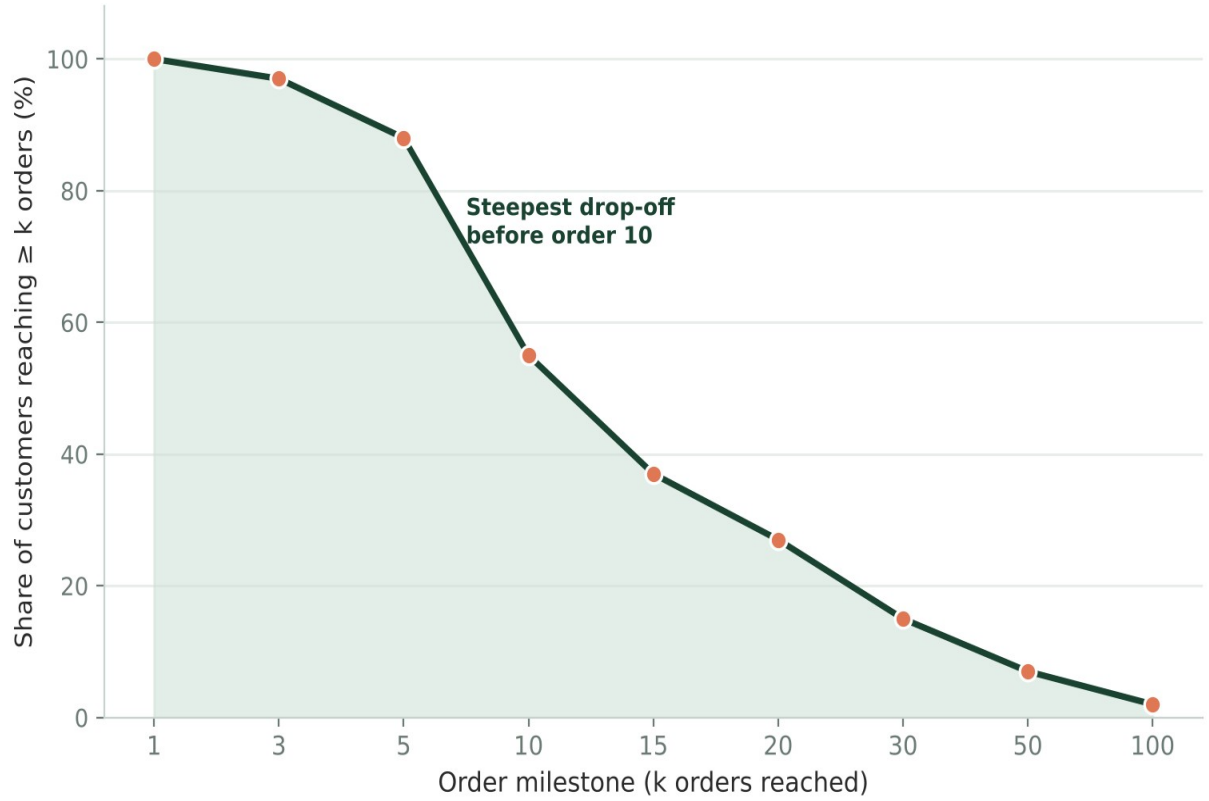


Sunday and Monday mornings see the heaviest order volume — a predictable window for staffing and promotions.

KEY INSIGHT 01

Early Orders Decide Who Stays

Customer retention by order milestone



Finding

- Customers who reach 10–15 orders show sharply higher long-term retention
- Weekly and biweekly reorder cycles form early in the lifecycle
- The steepest drop-off happens before the 10th order

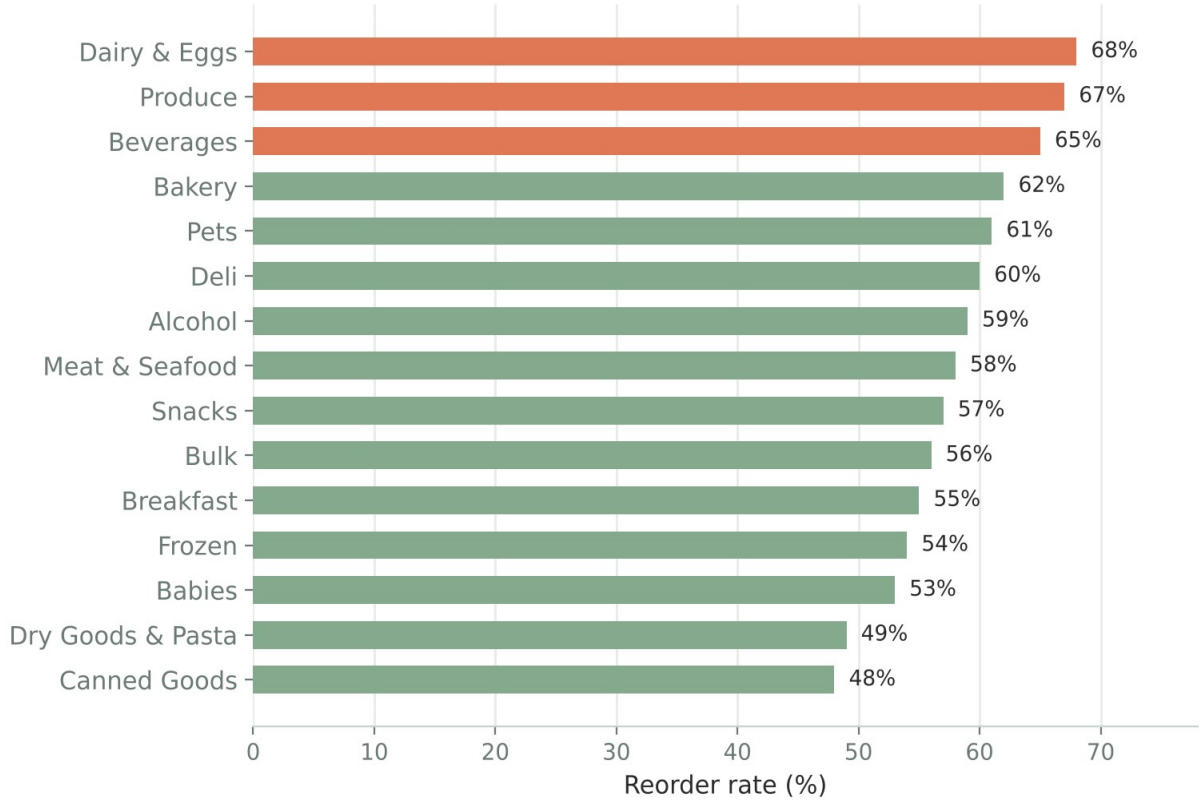
Business implication

Instacart gains the greatest return by focusing retention efforts on early-stage customers — not by trying to win back users after they've already disengaged.

KEY INSIGHT 02

Basket Diversity Predicts Staple Purchases

Reorder rate by product department



CART model result

- 78.3% accuracy predicting dairy/egg purchases from basket composition (vs. 66.6% baseline)
- Baskets spanning 4+ departments are far more likely to include staples like dairy and eggs
- Produce is bought often but reordered inconsistently — customers rotate items rather than repeat-buying the same one

Business implication

High-reorder categories suit auto-reorder prompts; low-repeat categories like produce benefit more from discovery-style recommendations.

CUSTOMER SEGMENTATION

Four Shopping Personas, Four Different Needs

01

Power Users

Frequent orders, high basket diversity, short gaps between orders. A disproportionate share of total order volume.

02

Stock-Up Shoppers

Moderate frequency but very large baskets and high monetary value — full grocery replacement behavior.

03

Routine Convenience

Order frequently but place small, low-diversity baskets — quick top-ups, not full trips.

04

Infrequent Shoppers

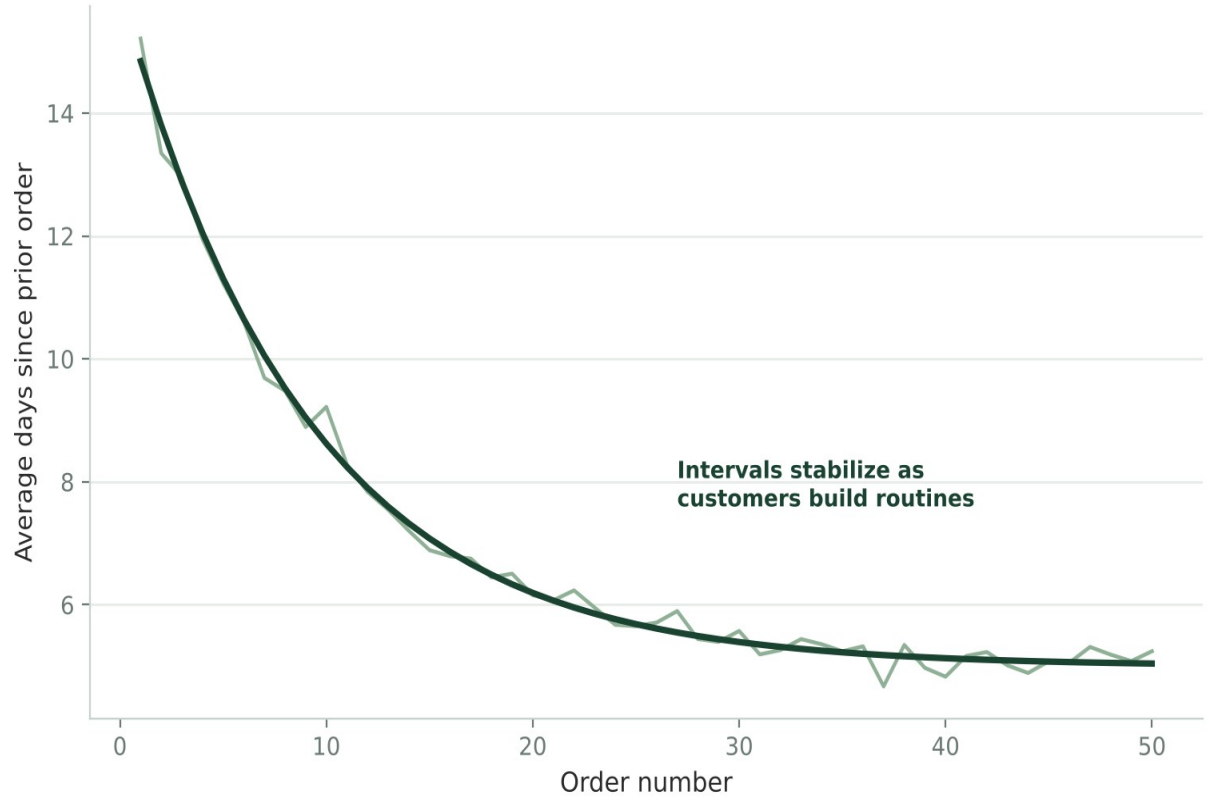
Low engagement, long gaps between orders, low reorder rates — highest churn risk.

Identified via K-Means clustering (k=4, confirmed by elbow method and silhouette analysis) on order frequency, recency, monetary value, basket size, and interpurchase time.

SYNTHESIS

Answering the Research Question

Order intervals stabilize as habits form



What drives retention & value

- Early habit formation in the first 10–15 orders
- Basket diversity and cross-category shopping behavior
- Which customer segment a shopper belongs to

Synthesis

Retention and value are driven by behavioral patterns, not single metrics like order count or frequency alone.

WHAT INSTACART SHOULD DO

Strategic Recommendations

Insight	Recommended action	Expected impact
Early churn, concentrated before order 10	Milestone-based onboarding nudges (reminders, incentives) through the first 10–15 orders	Higher early-stage retention
Low basket diversity limits staple purchases	Cross-category nudges at checkout — surface complementary departments during order creation	Larger, higher-value baskets
Segments differ sharply in behavior and value	Segment-specific promotions — subscriptions for Power Users, bundle deals for Stock-Up Shoppers	More efficient marketing spend

CONCLUSION

Small, Early Nudges Create Disproportionate Long-Term Value

- Higher customer retention and lifetime value
- Larger, more diverse, more valuable baskets
- More efficient, better-targeted marketing spend
- Stronger long-term customer relationships

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